SAI TEJA PEGALLAPATI

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Engineering Data Science Master's Student with enterprise experience designing scalable integration systems and implementing CI/CD-driven deployments. Strong foundation in machine learning, Python, and statistical modeling, complemented by hands-on expertise in Docker, Azure DevOps, and cloud-native architectures. Experienced in building reliable, production-grade systems that combine AI capabilities with DevOps automation and operational excellence.

Education

Master of Science, *Engineering Data Science* **May 2026**
University of Houston, TX
Bachelor of Technology, *Electronics and Communication Engineering* **June 2021**
Anurag Group of Institution, India

Skills

Tools and Languages Python, SQL, .NET, C, Java, HTML, CSS, JavaScript, Git, Linux, AWS, Azure DevOps, Kubernetes, Docker, Databricks, Tableau, Power BI, MATLAB, Visual Studio, TIBCO BW6/BW5, TIBCO EMS, TIBCO Administrator, TIBCO Hawk, REST/SOAP APIs, CI/CD Pipelines, Agile & Scrum

Technical Skills Machine Learning, Data Structures, Gen AI, Transformers, Embeddings, Probability & Statistics, Data Analytics & Visualization, Natural Language Processing, Data Mining, Engineering Analytics, Applied Neural Networks, Named Entity Recognition, Sequence Labelling, Prompt Engineering, Retrieval-Augmented Generation (RAG).

Professional Experience

Teaching Assistant **August 2025 -Present**
University of Houston *Houston, TX*

- Supported coursework and student projects in data structures, databases, and Python, focusing on data cleaning, ETL pipelines, SQL querying, debugging, and lab development.
- Collaborated with faculty to design and evaluate teaching materials, provided technical support and documentation.

System Engineer **August 2021 -July 2024**
Tata Consultancy Services *Hyderabad, IN*

- Designed and managed scalable TIBCO-based enterprise integrations across Dev, QA, UAT, and Prod environments.
- Built CI/CD pipelines using Azure DevOps, reducing deployment time by 30% and improving release consistency.
- Automated deployments with CLI and ANT scripts; integrated systems via REST, SOAP, JDBC, and JMS.
- Migrated TIBCO applications to Docker containers to enhance scalability and portability.
- Implemented infrastructure monitoring using TIBCO Hawk, Splunk, and custom log parsers.
- Boosted system reliability by 20% through proactive configuration management.
- Collaborated with cross-functional teams to enforce ITIL practices and ensure operational excellence.

Academic Projects

Anomaly Detection in Network Traffic (Deep Learning Project) **March 2025**
Python, PyOD, AutoEncoders, PCA, KDD Cup 1999 Dataset *University of Houston, TX*

- Built an unsupervised anomaly detection pipeline (KNN, Isolation Forest, AutoEncoder, VAE) on KDD Cup 1999; achieved >95% accuracy and 0.97 recall by optimizing reconstruction-error thresholds on normal-only training data. Applied PCA and ROC-AUC analysis to validate model performance and improve attack separability.

Concrete Strength Prediction Using Deep Neural Networks **December 2025**
TensorFlow/Keras, Regression Modeling, Materials Data *University of Houston, TX*

- Built a feedforward neural network to predict concrete compressive strength from 8 features, outperforming linear regression by capturing non-linear relationships and reducing MAE/RMSE.
- Improved model generalization using BatchNorm, Dropout, and adaptive learning rates; executed full ML pipeline from EDA to performance validation.

GAN-Based Medical Image Augmentation for Skin Lesion Classification **November 2025**

- Designed DCGAN and class-specific GAN architectures to generate 2,100 balanced synthetic dermoscopic images for the HAM10000 dataset, mitigating severe class imbalance and improving CNN classification accuracy to 79% through enhanced training diversity and robustness.

Certificates

AWS Certified Solutions Architect - Associate 2023
INTRO to Machine Learning: *Hewlett Packard Enterprise Data Science Institute* 2025
Introduction to Transformer-Based Natural Language Processing: *NVIDIA* 2025